PG-DAC · DATABASE TECHNOLOGIES · CCEE PRACTICE

DBT MCQ Bank

250 multiple-choice questions built from the Day 01–10 classwork: MySQL 8.4 and MongoDB. Every question shows the correct answer (highlighted in green) and an explanation. An answer key for quick self-testing is on the last page.

Output-based questions use the classwork tables (emp, dept, salgrade, emps, depts, meeting, emp_meeting). Their answers were run on MySQL 8.4.

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Easy 50 questions · Q1–Q50

Q1 SQL BASICS

CREATE, ALTER, DROP, TRUNCATE and RENAME belong to which category of SQL commands?

A. DCL
B. TCL
C. DDL
D. DML

Answer: C DDL (Data Definition Language) commands define or change the structure of database objects. They are auto-committed in MySQL.

Q2 SQL BASICS

Which of the following is a DML command?

A. GRANT
B. CREATE
C. INSERT
D. COMMIT

Answer: C DML (Data Manipulation Language) works on the rows of a table: INSERT, UPDATE, DELETE. CREATE is DDL, GRANT is DCL, COMMIT is TCL.

Q3 SQL BASICS

GRANT and REVOKE are examples of:

A. DDL
B. DQL
C. TCL
D. DCL

Answer: D DCL (Data Control Language) controls access rights. GRANT gives privileges and REVOKE takes them back.

Q4 SQL BASICS

COMMIT, ROLLBACK and SAVEPOINT belong to:

A. TCL
B. DCL
C. DML
D. DDL

Answer: A TCL (Transaction Control Language) commands manage transactions: make changes permanent (COMMIT), undo them (ROLLBACK) or mark points inside them (SAVEPOINT).

Q5 SQL BASICS

The SELECT statement belongs to which category?

A. DML
B. DCL
C. DDL
D. DQL

Answer: D SELECT only reads data, so it is classified as DQL (Data Query Language).

Q6 MYSQL CLIENT

Which command lists all databases visible to the current user?

A. SELECT DATABASES;
B. SHOW DATABASES;
C. LIST DATABASES;
D. SHOW TABLES;

Answer: B SHOW DATABASES; lists the databases the logged-in user can see. SHOW TABLES; lists tables of the currently selected database.

Q7 MYSQL CLIENT

Which command makes ac_classwork_db the current (default) database?

A. OPEN ac_classwork_db;
B. USE ac_classwork_db;
C. SELECT ac_classwork_db;
D. SET DATABASE ac_classwork_db;

Answer: B USE db_name; selects the database for the session. After that, unqualified table names refer to that database.

Q8 MYSQL CLIENT

Which function returns the name of the currently selected database?

A. VERSION()
B. SCHEMA_NAME()
C. DATABASE()
D. USER()

Answer: C DATABASE() (or its synonym SCHEMA()) returns the current database. USER() returns the logged-in user and VERSION() returns the server version.

Q9 DATA TYPES

Which string data type is fixed-length?

A. TEXT
B. VARCHAR
C. BLOB
D. CHAR

Answer: D CHAR(n) always reserves n characters and pads shorter values with trailing spaces. VARCHAR and TEXT are variable-length.

Q10 DATA TYPES

A column is declared as DECIMAL(9,2). What does it mean?

A. 9 bytes of storage and 2 decimal places
B. 9 digits in total, of which 2 are after the decimal point
C. Maximum value 9.2
D. 9 digits before the decimal point and 2 after

Answer: B In DECIMAL(M,D), M is the precision (total number of digits) and D is the scale (digits after the point). DECIMAL(9,2) can store up to 9999999.99.

Q11 DATA TYPES

What is the default format in which MySQL stores and displays a DATE value?

A. DD-MON-YY
B. DD-MM-YYYY
C. MM/DD/YYYY
D. YYYY-MM-DD

Answer: D MySQL uses the ISO format 'YYYY-MM-DD' for DATE, e.g. '1981-11-17'.

Q12 DATA TYPES

Which data type is best suited to store a photograph (raw binary data) inside a table?

A. ENUM
B. TEXT
C. BLOB
D. VARCHAR

Answer: C BLOB (Binary Large Object) types store binary data such as images or files. TEXT is for large character data.

Q13 OPERATORS

In a LIKE pattern, which wildcard matches exactly one character?

A. _
B. ?
C. %
D. *

Answer: A The underscore (_) matches exactly one character. The percent sign (%) matches zero or more characters.

Q14 OPERATORS

Which names does the condition ename LIKE 'M%' return?

A. Names containing M anywhere
B. Names ending with M
C. Names whose second letter is M
D. Names starting with M (including the name 'M' itself)

Answer: D % means zero or more characters, so 'M%' matches every value that starts with M, including a value that is just 'M'.

Q15 OPERATORS

WHERE sal BETWEEN 1000 AND 2000 is equivalent to:

A. sal > 1000 AND sal <= 2000
B. sal > 1000 AND sal < 2000
C. sal >= 1000 OR sal <= 2000
D. sal >= 1000 AND sal <= 2000

Answer: D BETWEEN is inclusive on both ends: low <= value <= high.

Q16 NULL

Which is the correct way to find employees who do not earn a commission (comm is NULL)?

A. WHERE comm == NULL
B. WHERE comm IS NULL
C. WHERE comm = 0
D. WHERE comm = NULL

Answer: B NULL cannot be compared with = because any comparison with NULL gives NULL (unknown). Use IS NULL / IS NOT NULL.

Q17 SELECT

What does the DISTINCT keyword do in SELECT DISTINCT job FROM emp?

A. Removes NULL values only
B. Removes duplicate rows from the result
C. Counts the jobs
D. Sorts the jobs

Answer: B DISTINCT eliminates duplicate rows in the result set. With several columns it keeps unique combinations.

Q18 SELECT

If ORDER BY sal is written without ASC or DESC, the rows are sorted in:

A. Random order
B. Ascending order
C. Descending order
D. Insertion order

Answer: B ASC is the default sort order for ORDER BY.

Q19 SELECT

SELECT * FROM emp LIMIT 5; returns:

A. The last 5 rows
B. 5 random rows
C. The first 5 rows of the result
D. Rows 6 onwards

Answer: C LIMIT n returns the first n rows of the result (after ORDER BY, if one is given).

Q20 DDL

Which command removes the table structure along with all its data?

A. DROP
B. DELETE
C. TRUNCATE
D. REMOVE

Answer: A DROP TABLE removes the data, the structure, its indexes and triggers. TRUNCATE removes only the rows and keeps the structure.

Q21 DDL

TRUNCATE TABLE is classified as:

A. DDL
B. TCL
C. DCL
D. DML

Answer: A TRUNCATE is a DDL command. It drops and recreates the table internally, so it is auto-committed and cannot be rolled back.

Q22 DML

What happens when you run DELETE FROM emp; (with no WHERE clause)?

A. An error is raised
B. Only the first row is deleted
C. All rows are deleted; the table structure remains
D. The table is dropped

Answer: C Without WHERE, DELETE removes every row but the table definition stays. Inside a transaction it can be rolled back.

Q23 MYSQL CLIENT

DESC emp; is used to:

A. Describe the data in emp
B. Delete the emp table
C. Display the structure (columns and data types) of emp
D. Sort emp in descending order

Answer: C DESC (DESCRIBE) shows each column with its data type, NULL-ability, key and default.

Q24 AGGREGATE FUNCTIONS

COUNT(comm) counts:

A. Only rows where comm is NULL
B. Only rows where comm is NOT NULL
C. Distinct values including NULL
D. All rows including NULLs

Answer: B Group functions ignore NULL. COUNT(column) counts non-NULL values; COUNT(*) counts all rows.

Q25 AGGREGATE FUNCTIONS

The emp table has 14 rows. What does SELECT COUNT(*) FROM emp; return?

A. 13
B. 14
C. 4
D. 1

Answer: B COUNT(*) counts every row regardless of NULLs, so it returns 14.

Q26 GROUP BY

Which clause filters groups produced by GROUP BY?

A. HAVING
B. LIMIT
C. ORDER BY
D. WHERE

Answer: A WHERE filters individual rows before grouping. HAVING filters the groups after grouping, so it can use group functions.

Q27 FUNCTIONS

Which function returns the current date and time?

A. NOW()
B. DATE()
C. CURTIME()
D. CURDATE()

Answer: A NOW() returns 'YYYY-MM-DD HH:MM:SS'. CURDATE() returns only the date and CURTIME() only the time.

Q28 FUNCTIONS

What is the output of SELECT UPPER('sunbeam');?

A. sunbeam
B. Sunbeam
C. SUNBEAM
D. sUNBEAM

Answer: C UPPER() converts every character to upper case. LOWER() does the reverse.

Q29 FUNCTIONS

What does SELECT LENGTH('Sunbeam'); return?

A. NULL
B. 8
C. 6
D. 7

Answer: D 'Sunbeam' has 7 characters. LENGTH() returns the length in bytes, which equals 7 for plain ASCII text.

Q30 FUNCTIONS

What does SELECT CONCAT('DB','T'); return?

A. DBT
B. DB+T
C. DB T
D. An error

Answer: A CONCAT() joins its arguments into one string with no separator.

Q31 CONSTRAINTS

A PRIMARY KEY column is:

A. Always auto-incremented
B. Unique and NOT NULL
C. Not unique but NOT NULL
D. Unique and may contain NULL

Answer: B A primary key identifies each row uniquely, so it combines UNIQUE and NOT NULL. AUTO_INCREMENT is optional.

Q32 CONSTRAINTS

How many PRIMARY KEY constraints can a table have?

A. Any number
B. Only one (it may cover several columns)
C. One per column
D. Two

Answer: B A table can have only one primary key. That key may be composite, i.e. made of more than one column.

Q33 CONSTRAINTS

A FOREIGN KEY in the child table normally refers to which column of the parent table?

A. Its first column
B. Its indexed TEXT column
C. Its PRIMARY KEY (or a UNIQUE key)
D. Any column

Answer: C A foreign key must reference a key that uniquely identifies parent rows: the primary key or a unique key.

Q34 JOINS

Which join returns only the rows that have matching values in both tables?

A. RIGHT JOIN
B. LEFT JOIN
C. CROSS JOIN
D. INNER JOIN

Answer: D INNER JOIN keeps only rows where the join condition is satisfied in both tables.

Q35 JOINS

A CROSS JOIN between a table of 4 rows and a table of 5 rows returns how many rows?

A. 9
B. 4
C. 5
D. 20

Answer: D A CROSS JOIN is the Cartesian product: every row of the first table pairs with every row of the second, 4 x 5 = 20.

Q36 JOINS

A SELF JOIN is:

A. A join of a table with itself
B. A join of a view with a table
C. A join on all columns with the same name
D. A join without a condition

Answer: A In a self join the same table appears twice with different aliases, e.g. emp e JOIN emp m ON e.mgr = m.empno.

Q37 SET OPERATORS

What is the difference between UNION and UNION ALL?

A. UNION removes duplicates; UNION ALL keeps them
B. UNION ALL removes duplicates; UNION keeps them
C. No difference
D. UNION sorts columns; UNION ALL does not

Answer: A UNION returns distinct rows from both queries. UNION ALL returns every row, including duplicates, and is faster.

Q38 VIEWS

A view is:

A. A virtual table defined by a stored SELECT query
B. A backup of the database
C. An index on a table
D. A physical copy of a table

Answer: A A view stores only the query. Each time it is used, the query runs against the base tables.

Q39 INDEXES

The main purpose of creating an index is to:

A. Speed up searching and retrieval of rows
B. Encrypt data
C. Enforce foreign keys
D. Save disk space

Answer: A An index lets the server find rows without scanning the whole table. It speeds up SELECT but adds overhead to INSERT/UPDATE/DELETE.

Q40 TRANSACTIONS

Which command makes all changes of the current transaction permanent?

A. END
B. COMMIT
C. ROLLBACK
D. SAVE

Answer: B COMMIT ends the transaction and saves its changes permanently. ROLLBACK ends it and discards the changes.

Q41 TRANSACTIONS

In the ACID properties, 'A' stands for:

A. Accuracy
B. Authorization
C. Availability
D. Atomicity

Answer: D Atomicity: all operations of a transaction succeed together or none of them takes effect.

Q42 STORED PROCEDURES

Which statement executes a stored procedure in MySQL?

A. CALL proc();
B. RUN proc();
C. EXECUTE proc;
D. EXEC proc();

Answer: A CALL procedure_name(args); runs a stored procedure in MySQL. Stored functions are used inside expressions such as SELECT fn(x);

Q43 STORED PROCEDURES

If no mode is written before a stored procedure parameter, the parameter is:

A. INOUT
B. OUT
C. Invalid
D. IN

Answer: D IN is the default mode. It only passes a value into the procedure.

Q44 TRIGGERS

A trigger is:

A. A procedure that must be called with CALL
B. A stored program that runs automatically on INSERT, UPDATE or DELETE of a table
C. A type of index
D. A scheduled job

Answer: B Triggers are fired automatically by DML events (BEFORE/AFTER INSERT, UPDATE, DELETE) on the table they belong to.

Q45 NORMALIZATION

The main goal of normalization is to:

A. Encrypt tables
B. Reduce data redundancy and avoid insert/update/delete anomalies
C. Create more indexes
D. Increase redundancy for faster reads

Answer: B Normalization splits data into well-structured tables so each fact is stored once, which removes anomalies.

Q46 NORMALIZATION

A table is in First Normal Form (1NF) when:

A. It has no partial dependency
B. It has a foreign key
C. Every column holds atomic (single) values and there are no repeating groups
D. It has no transitive dependency

Answer: C 1NF requires atomic values: no lists, multi-valued attributes or repeating groups in a column.

Q47 MONGODB

MongoDB stores records as:

A. XML files
B. Key-value pairs only
C. Rows in tables
D. BSON documents

Answer: D MongoDB is a document database. Documents are JSON-like and stored in binary JSON (BSON).

Q48 MONGODB

In MongoDB, the equivalent of an RDBMS table is a:

A. Cursor
B. Field
C. Collection
D. Document

Answer: C A database holds collections, a collection holds documents, and a document holds fields. Collection ~ table, document ~ row, field ~ column.

Q49 MONGODB

Which MongoDB method inserts several documents in one call?

A. db.emp.insertOne()
B. db.emp.push()
C. db.emp.insertMany()
D. db.emp.add()

Answer: C insertMany() takes an array of documents. insertOne() inserts a single document.

Q50 NOSQL

The term 'NoSQL' is commonly expanded as:

A. Not Only SQL
B. No SQL allowed
C. New Object SQL
D. Non-Structured Query Language

Answer: A 'Not Only SQL' describes databases that use models other than relational tables: document, key-value, column-family and graph.

Moderate 100 questions · Q51–Q150

Q51 NUMERIC FUNCTIONS

What is the output?

```
SELECT ROUND(12345.567, -2);
```

A. 12345.57
B. 12400
C. 12300
D. 12345.00

Answer: C A negative second argument rounds to the left of the decimal point. -2 rounds to the nearest hundred: 12345.567 becomes 12300.

Q52 NUMERIC FUNCTIONS

What is the output?

```
SELECT TRUNCATE(12456.8467, 2);
```

A. 12456.84
B. 12456
C. 12500
D. 12456.85

Answer: A TRUNCATE(x, d) cuts the number to d decimal places without rounding, giving 12456.84.

Q53 NUMERIC FUNCTIONS

What is the output?

```
SELECT CEIL(-4.3), FLOOR(-4.3);
```

A. -4, -5
B. -4, -4
C. -5, -5
D. -5, -4

Answer: A CEIL returns the smallest integer >= x (-4). FLOOR returns the largest integer <= x (-5). For negative numbers, 'up' moves towards zero.

Q54 STRING FUNCTIONS

What is the output?

```
SELECT LPAD('Sunbeam', 10, '*');
```

A. *Sunbeam**
B. Sunbeam***
C. **********Sunbeam
D. ***Sunbeam

Answer: D LPAD pads on the left until the total length is 10. 'Sunbeam' has 7 characters, so 3 '*' are added on the left.

Q55 STRING FUNCTIONS

What is the output?

```
SELECT SUBSTRING('Sunbeam Info', 5);
```

A. eam
B. Sunb
C. beam Info
D. eam Info

Answer: D SUBSTRING(str, pos) starts at position 5 (1-based) and returns everything to the end: 'eam Info'.

Q56 STRING FUNCTIONS

What is the output?

```
SELECT SUBSTRING('Sunbeam Info', 5, 3);
```

A. am
B. eam
C. eam Info
D. bea

Answer: B SUBSTRING(str, pos, len) returns 3 characters starting at position 5: 'e','a','m'.

Q57 STRING FUNCTIONS

What is the output?

```
SELECT LEFT('Sunbeam',3), RIGHT('Sunbeam',3);
```

A. Sunb, beam
B. Sun, eam
C. Sun, mae
D. nuS, eam

Answer: B LEFT returns the first n characters and RIGHT the last n characters, in their original order.

Q58 NULL

For employee SMITH, sal = 800 and comm = NULL. What does sal + comm return?

A. An error
B. NULL
C. 800
D. 0

Answer: B Any arithmetic with NULL gives NULL. Use IFNULL(comm,0) or COALESCE(comm,0) to treat a missing commission as zero.

Q59 FLOW FUNCTIONS

What is the output?

```
SELECT NULLIF(10, 10), NULLIF(10, 20);
```

A. 10, 10
B. NULL, NULL
C. NULL, 10
D. 0, 10

Answer: C NULLIF(a,b) returns NULL when a = b, otherwise it returns a. A common use is avoiding division by zero: sales / NULLIF(orders, 0).

Q60 FLOW FUNCTIONS

What is the output?

```
SELECT COALESCE(NULL, NULL, 5, 10);
```

A. 10
B. NULL
C. 5
D. 15

Answer: C COALESCE returns the first non-NULL value in its list of arguments.

Q61 DATE FUNCTIONS

What is the output?

```
SELECT DATEDIFF('2024-03-01', '2024-02-01');
```

A. 28
B. 29
C. 1
D. 30

Answer: B DATEDIFF returns (date1 - date2) in days. February 2024 is a leap-year month with 29 days.

Q62 DATE FUNCTIONS

What is the output?

```
SELECT TIMESTAMPDIFF(MONTH, '2009-01-12', '2010-12-15');
```

A. 23
B. 22
C. 11
D. 24

Answer: A From 2009-01-12 to 2010-12-12 is 23 complete months; the 3 extra days do not make another full month.

Q63 DATE FUNCTIONS

What is the output?

```
SELECT DATE_ADD('2023-12-31 23:59:59', INTERVAL 1 SECOND);
```

A. 2023-12-31 24:00:00
B. 2023-12-31 00:00:00
C. 2024-01-01 00:00:00
D. An error

Answer: C Adding one second rolls over the minute, hour, day, month and year, giving midnight of 1 January 2024.

Q64 DATE FUNCTIONS

Which call formats hire = '1982-01-05' as 'January 5th 1982, Tuesday'?

A. TO_CHAR(hire, 'Month DDth YYYY, Day')
B. DATE_FORMAT(hire, '%m %d %y, %w')
C. DATE_FORMAT(hire, '%M %D %Y, %W')
D. DATE_FORMAT(hire, '%b %e %Y, %a')

Answer: C %M = full month name, %D = day with English suffix (5th), %Y = 4-digit year, %W = full weekday name. Lower-case specifiers give numbers.

Q65 AGGREGATE FUNCTIONS

comm values in emp are 300, 500, 1400, 0 and NULL for the other 10 rows. What is AVG(comm)?

```
SELECT AVG(comm) FROM emp;
```

A. 550.00
B. 2200.00
C. NULL
D. 157.14

Answer: A AVG ignores NULLs. It averages only the 4 non-NULL values: (300+500+1400+0)/4 = 550.

Q66 AGGREGATE FUNCTIONS

Using the same data, what does this return?

```
SELECT AVG(IFNULL(comm, 0)) FROM emp;
```

A. NULL
B. 550.000000
C. 157.142857
D. 0

Answer: C IFNULL turns every NULL into 0, so all 14 rows take part: 2200/14 = 157.142857.

Q67 AGGREGATE FUNCTIONS

What is the output?

```
SELECT COUNT(*), COUNT(comm), COUNT(DISTINCT job) FROM emp;
```

A. 14, 4, 14
B. 14, 14, 5
C. 14, 4, 5
D. 4, 4, 5

Answer: C COUNT(*) = 14 rows; COUNT(comm) = 4 non-NULL commissions; the distinct jobs are CLERK, SALESMAN, MANAGER, ANALYST, PRESIDENT = 5.

Q68 GROUP BY

What is the total salary of department 10 (KING 5000, CLARK 2450, MILLER 1300)?

```
SELECT SUM(sal) FROM emp WHERE deptno = 10;
```

A. 7450
B. 2916.67
C. 8700
D. 8750

Answer: D 5000 + 2450 + 1300 = 8750.

Q69 GROUP BY

Which jobs does this query return?

```
SELECT job, COUNT(*) FROM emp GROUP BY job HAVING COUNT(*) >= 3;
```

A. CLERK and SALESMAN
B. ANALYST and PRESIDENT
C. CLERK, SALESMAN and MANAGER
D. MANAGER only

Answer: C CLERK has 4, SALESMAN has 4, MANAGER has 3, ANALYST has 2 and PRESIDENT has 1. Three groups pass HAVING COUNT(*) >= 3.

Q70 GROUP BY

With ONLY_FULL_GROUP_BY enabled (the MySQL 8 default), what happens here?

```
SELECT ename, MAX(sal) FROM emp;
```

A. Returns KING, 5000

B. Error 1140: non-aggregated column in aggregated query

C. Returns 14 rows

D. Returns the first ename with 5000

Answer: B A normal column cannot be selected with a group function unless it is in GROUP BY. MySQL raises ERROR 1140 under only_full_group_by.

Q71 GROUP BY

What is the result of this query?

```
SELECT * FROM emp WHERE sal = MAX(sal);
```

A. Empty set

B. All 14 rows

C. The row of KING

D. ERROR 1111: Invalid use of group function

Answer: D Group functions cannot be used in the WHERE clause, because WHERE is evaluated before grouping. Use a subquery: WHERE sal = (SELECT MAX(sal) FROM emp).

Q72 GROUP BY

Which statement about GROUP BY is TRUE?

A. Every non-aggregated column in SELECT must appear in GROUP BY

B. GROUP BY must come after HAVING

C. GROUP BY cannot be used with WHERE

D. Every column in GROUP BY must appear in SELECT

Answer: A Columns in GROUP BY need not appear in SELECT, but every plain (non-aggregated) column in SELECT must be part of GROUP BY.

Q73 QUERY EXECUTION

What is the logical order in which the clauses of a SELECT are processed?

A. SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY

B. FROM, WHERE, GROUP BY, HAVING, SELECT, ORDER BY, LIMIT

C. FROM, GROUP BY, WHERE, SELECT, HAVING, ORDER BY

D. WHERE, FROM, SELECT, GROUP BY, ORDER BY, HAVING

Answer: B Rows come from FROM, are filtered by WHERE, grouped by GROUP BY, groups are filtered by HAVING, columns are computed in SELECT, then ORDER BY and LIMIT apply.

Q74 SELECT

Why does this query fail with 'Unknown column allowance'?

```
SELECT ename, sal*0.3 AS allowance FROM emp WHERE allowance > 500;
```

A. A column alias from SELECT cannot be used in WHERE, because WHERE is evaluated before SELECT

B. sal*0.3 is not allowed in SELECT

C. Aliases need double quotes

D. AS is not supported in MySQL

Answer: A WHERE runs before the SELECT list, so the alias does not exist yet. Repeat the expression (WHERE sal*0.3 > 500). Aliases are allowed in ORDER BY.

Q75 SELECT

What does ORDER BY 3 DESC do in this query?

```
SELECT empno, ename, sal, job FROM emp ORDER BY 3 DESC;
```

A. Syntax error

B. Returns only 3 rows

C. Sorts by the 3rd column of the table

D. Sorts by sal (3rd column of the SELECT list) in descending order

Answer: D A number in ORDER BY refers to the position of the column in the SELECT list, so 3 means sal.

Q76 SELECT

What does LIMIT 5, 6 return?

```
SELECT * FROM emp LIMIT 5, 6;
```

A. Rows 5 to 11 inclusive

B. Rows 5 to 6

C. 5 rows starting from row 6

D. 6 rows after skipping the first 5 rows

Answer: D LIMIT offset, count: the first number is how many rows to skip, the second is how many to return. Rows 6 to 11 are returned.

Q77 OPERATORS

How many employees have exactly 4 characters in their name?

```
SELECT ename FROM emp WHERE ename LIKE '____';
```

A. 3

B. 4

C. 2

D. 5

Answer: A Four underscores match exactly four characters: WARD, KING and FORD.

Q78 OPERATORS

Which names are returned?

```
SELECT ename FROM emp WHERE ename LIKE '_A%';
```

A. WARD, MARTIN, JAMES
B. ALLEN, ADAMS
C. BLAKE, CLARK
D. WARD, JAMES

Answer: A '_A%' means any one character, then 'A' as the second character, then anything. WARD, MARTIN and JAMES match.

Q79 OPERATORS

Why are MARTIN and MILLER NOT returned here?

```
SELECT ename FROM emp WHERE ename BETWEEN 'C' AND 'M';
```

A. 'MARTIN' is greater than 'M' in string comparison, so it lies outside the range
B. Because of case sensitivity
C. BETWEEN is exclusive
D. LIKE should be used for strings

Answer: A Strings compare character by character. 'M' is a prefix of 'MARTIN', and the longer string is greater, so 'MARTIN' > 'M'. Use ename < 'N' or LEFT(ename,1) BETWEEN 'C' AND 'M'.

Q80 OPERATORS

What does this query return?

```
SELECT * FROM emp WHERE comm = NULL;
```

A. An error
B. The 10 employees without commission
C. The 4 employees with commission
D. Empty set

Answer: D comm = NULL evaluates to NULL (unknown) for every row, and WHERE keeps only TRUE rows. Use comm IS NULL.

Q81 OPERATORS

Which condition is equivalent to job NOT IN ('CLERK','SALESMAN')?

A. NOT job = 'CLERK' OR job = 'SALESMAN'
B. job <> 'CLERK' AND job <> 'SALESMAN'
C. job <> 'CLERK' OR job <> 'SALESMAN'
D. job != ALL('CLERK') OR 'SALESMAN'

Answer: B NOT IN means 'different from every value in the list', i.e. the conditions are joined with AND.

Q82 CASE

Which CASE expression is syntactically correct?

A. CASE WHEN deptno = 10 'ACCOUNTING' END
B. CASE deptno IF 10 THEN 'ACCOUNTING' END CASE
C. CASE deptno WHEN 10 THEN 'ACCOUNTING' WHEN 20 THEN 'RESEARCH' ELSE 'OTHER' END
D. SWITCH deptno CASE 10 : 'ACCOUNTING'

Answer: C Simple CASE: CASE expr WHEN value THEN result ... ELSE default END. Searched CASE: CASE WHEN condition THEN result ... END. END CASE is used only for the CASE statement inside stored programs.

Q83 DATA TYPES

In strict SQL mode, what happens on this INSERT?

```
CREATE TABLE temp(col1 CHAR(4));
INSERT INTO temp VALUES ('abcde');
```

A. Stored as 'abcd'
B. Stored as 'abcde'
C. Stored as NULL
D. ERROR 1406: Data too long for column 'col1'

Answer: D In strict mode (the default in MySQL 8), a value longer than the column size is rejected with ERROR 1406 instead of being silently truncated.

Q84 DDL VS DML

Which statement about DELETE and TRUNCATE is TRUE?

A. DELETE resets AUTO_INCREMENT; TRUNCATE does not
B. Both are DML
C. DELETE can use WHERE and can be rolled back inside a transaction; TRUNCATE cannot
D. TRUNCATE can use WHERE

Answer: C DELETE is DML: row-by-row, WHERE allowed, and it can be rolled back. TRUNCATE is DDL: no WHERE, auto-committed, and it resets AUTO_INCREMENT.

Q85 ALTER

Which statement adds a column ph_number right after ename?

A. ALTER TABLE emp ADD ph_number CHAR(10) AFTER ename;
B. UPDATE TABLE emp ADD ph_number AFTER ename;
C. ALTER emp ADD COLUMN ph_number NEXT ename;
D. ALTER TABLE emp INSERT ph_number CHAR(10) AFTER ename;

Answer: A ALTER TABLE ... ADD [COLUMN] col type [FIRST | AFTER col]. Without FIRST/AFTER the column is added at the end.

Q86 ALTER

Which statement renames column ename to first_name without changing its type (MySQL 8)?

A. ALTER TABLE emp RENAME COLUMN ename TO first_name;
B. ALTER TABLE emp RENAME ename TO first_name;
C. ALTER TABLE emp MODIFY ename first_name;
D. RENAME COLUMN emp.ename TO first_name;

Answer: A MySQL 8 supports RENAME COLUMN old TO new. CHANGE old new type also works but requires the full column definition.

Q87 ALTER

Which ALTER clause changes only the data type of an existing column, e.g. ename to VARCHAR(50)?

A. ALTER TABLE emp UPDATE ename VARCHAR(50);
B. ALTER TABLE emp RETYPE ename VARCHAR(50);
C. ALTER TABLE emp MODIFY ename VARCHAR(50);
D. ALTER TABLE emp ALTER ename VARCHAR(50);

Answer: C MODIFY redefines a column while keeping its name. CHANGE can rename and redefine at the same time.

Q88 REGEXP

Which values match col1 REGEXP '^is'?

A. Values equal to 'is' only
B. Values that contain 'is' anywhere
C. Values that start with 'is', such as 'isnt' and 'is'
D. Values that end with 'is'

Answer: C ^ anchors the pattern to the beginning of the string. 'is$' would anchor it to the end.

Q89 REGEXP

From bag, beg, big, bog, bug, b*g, bg, which values match the pattern 'b[aiu]g'?

A. bag, big, bug
B. All except bg
C. beg, bog
D. bag, beg, big

Answer: A [aiu] matches exactly one character that is a, i or u.

Q90 REGEXP

From bag, beg, big, bog, bug, b*g, bg, xyz, which value matches 'b[^a-z]g'?

A. bg
B. xyz
C. bag
D. b*g

Answer: D [^a-z] matches exactly one character that is NOT a letter. Only '*' satisfies it. 'bg' fails because one character is required between b and g.

Q91 REGEXP

In MySQL regular expressions, what does '+' mean?

A. Exactly one occurrence
B. 1 or more occurrences of the previous character
C. Concatenation
D. 0 or 1 occurrence of the previous character

Answer: B * = 0 or more, + = 1 or more, ? = 0 or 1, {n} = exactly n, {m,n} = between m and n occurrences.

Q92 ROLLUP

What does GROUP BY deptno WITH ROLLUP add to the result?

A. A duplicate of every group
B. Sorted output only
C. A row per employee
D. A grand-total row, shown with NULL in the deptno column

Answer: D WITH ROLLUP adds super-aggregate rows. For a single grouping column it adds one grand-total row in which deptno is NULL.

Q93 ROLLUP

What does GROUPING(deptno) return in a WITH ROLLUP query?

A. 1 for super-aggregate (subtotal or total) rows where deptno is rolled up, else 0
B. The number of rows in the group
C. The deptno value
D. NULL for normal rows

Answer: A GROUPING() tells a real NULL value apart from the NULL produced by ROLLUP. It returns 1 on rows generated by ROLLUP.

Q94 JOINS

emps has deptno values 10, 10, 20, 50, 50; depts has 10, 20, 30, 40. How many rows does the INNER JOIN return?

```
SELECT e.ename, d.dname FROM emps e INNER JOIN depts d ON e.deptno = d.deptno;
```

A. 3
B. 20
C. 4
D. 5

Answer: A Only Amit (10), Rahul (10) and Nilesh (20) have a matching deptno. Nitin and Sarang are in dept 50, which does not exist in depts.

Q95 JOINS

With the same emps/depts data, how many rows does emps LEFT JOIN depts return?

A. 7
B. 4
C. 3
D. 5

Answer: D A LEFT JOIN keeps every row of the left table (5 emps). Non-matching rows get NULL in the depts columns.

Q96 JOINS

With the same data, how many rows does emps RIGHT JOIN depts return?

A. 7
B. 3
C. 5
D. 4

Answer: C Dept 10 matches 2 emps, dept 20 matches 1, and depts 30 and 40 appear once each with NULL emps: 2+1+1+1 = 5.

Q97 JOINS

How is a FULL OUTER JOIN produced in MySQL?

A. FULL OUTER JOIN keyword
B. LEFT JOIN ... UNION ... RIGHT JOIN
C. CROSS JOIN with WHERE
D. NATURAL JOIN

Answer: B MySQL has no FULL OUTER JOIN. Combine a LEFT JOIN and a RIGHT JOIN with UNION, which also removes the duplicated matching rows.

Q98 JOINS

What does JOIN depts d USING (deptno) do?

A. Creates a cross join
B. Joins on every column with the same name
C. Joins on e.deptno = d.deptno, and deptno appears only once in SELECT *
D. Requires deptno to be a primary key

Answer: C USING(col) is shorthand for an equi-join on a column that has the same name in both tables; the column is merged in the output.

Q99 JOINS

Which is an example of a NON-equi join?

A. emp e JOIN dept d ON e.deptno = d.deptno
B. emp e JOIN emp m ON e.mgr = m.empno
C. emp e JOIN salgrade s ON e.sal BETWEEN s.losal AND s.hisal
D. emp NATURAL JOIN dept

Answer: C A non-equi join uses an operator other than = (BETWEEN, <, >, !=). Matching a salary to a grade range is the classic example.

Q100 JOINS

To list every employee with the name of his/her manager, the correct join condition is:

```
SELECT e.ename, m.ename AS manager FROM emp e JOIN emp m ON ____;
```

A. e.empno = m.mgr
B. e.empno = m.empno
C. e.mgr = m.empno
D. e.mgr = m.mgr

Answer: C e is the employee and m plays the manager. The employee's mgr column holds the manager's empno.

Q101 CONSTRAINTS

A column has a UNIQUE constraint. Which INSERT sequence succeeds?

A. Neither; NULL is not allowed
B. Only if it is also NOT NULL
C. Two rows with the same non-NULL value
D. Two rows that both have NULL in that column

Answer: D MySQL allows any number of NULLs in a UNIQUE column, because NULL is not equal to anything, including another NULL.

Q102 CONSTRAINTS

Where must a composite PRIMARY KEY be declared?

A. At table level, e.g. PRIMARY KEY(roll_no, std)
B. In a separate ALTER only
C. At column level
D. It cannot be composite

Answer: A A constraint that covers more than one column can only be written at table level.

Q103 CONSTRAINTS

Which constraint can be declared ONLY at column level?

A. UNIQUE
B. CHECK
C. NOT NULL
D. PRIMARY KEY

Answer: C NOT NULL is a column-level constraint. UNIQUE, PRIMARY KEY, FOREIGN KEY and CHECK can also be written at table level and given names.

Q104 CONSTRAINTS

Which error appears when a duplicate value is inserted into a PRIMARY KEY or UNIQUE column?

A. ERROR 3819
B. ERROR 1062: Duplicate entry
C. ERROR 1048
D. ERROR 1452

Answer: B 1062 = duplicate key. 1048 = column cannot be null. 1452 = foreign key fails. 3819 = check constraint violated.

Q105 CONSTRAINTS

ERROR 1452 'Cannot add or update a child row: a foreign key constraint fails' occurs when:

A. Two foreign keys are declared
B. A child row refers to a parent key value that does not exist
C. The parent row is deleted
D. A NULL is inserted in a primary key

Answer: B The foreign key value in the child table must exist in the referenced parent column, or be NULL.

Q106 CONSTRAINTS

With FOREIGN KEY (deptno) REFERENCES dept(deptno) ON DELETE CASCADE, deleting dept 10 will:

A. Fail because employees exist
B. Set deptno to NULL for those employees
C. Delete only dept 10
D. Delete dept 10 and all employees of dept 10

Answer: D CASCADE propagates the parent delete to the matching child rows. ON DELETE SET NULL would set the child foreign key to NULL instead.

Q107 CONSTRAINTS

Why would a DBA run SET FOREIGN_KEY_CHECKS = 0;?

A. To temporarily disable foreign key validation, for example during data migration
B. To delete all foreign keys
C. To speed up SELECT queries
D. To make foreign keys unique

Answer: A It switches off referential checks for the session so data can be loaded in any order. It must be set back to 1 afterwards.

Q108 CONSTRAINTS

An INSERT violates CHECK (sal > 0). What happens in MySQL 8?

A. ERROR 3819: Check constraint is violated
B. The row is inserted with a warning
C. The value is set to 0
D. CHECK is ignored by MySQL

Answer: A Since MySQL 8.0.16, CHECK constraints are enforced. Older versions parsed but ignored them.

Q109 INDEXES

When you create a UNIQUE constraint on a column, MySQL internally:

A. Creates a view
B. Does nothing extra
C. Creates a trigger
D. Creates a unique index on that column

Answer: D UNIQUE and PRIMARY KEY constraints are implemented with unique indexes. You can see them with SHOW INDEXES FROM table;

Q110 INDEXES

In EXPLAIN output, access_type = "ALL" means:

A. An index-only scan
B. A full table scan
C. All indexes are used
D. All rows are cached

Answer: B ALL is a full table scan. After adding a suitable index the access type usually becomes ref or range and query_cost drops.

Q111 VIEWS

Which kind of view is generally NOT updatable (INSERT/UPDATE/DELETE fail)?

A. A view with a WHERE clause
B. A view with column aliases
C. A view on one table with selected columns
D. A view that uses GROUP BY / aggregate functions

Answer: D A view with GROUP BY, DISTINCT, aggregates, UNION and similar constructs has no one-to-one mapping to base rows, so DML on it fails.

Q112 VIEWS

What does WITH CHECK OPTION do on a view defined as SELECT * FROM emp WHERE comm IS NULL?

A. Adds a CHECK constraint to emp
B. Prevents SELECT on the view
C. Checks the syntax of the view
D. Rejects INSERT/UPDATE through the view if the resulting row would not satisfy comm IS NULL

Answer: D Without it, a row inserted through the view with a commission goes into emp but disappears from the view. WITH CHECK OPTION blocks this with ERROR 1369.

Q113 SUBQUERIES

A single-row subquery can be used with which operators?

A. BETWEEN only
B. =, <>, <, >, <=, >=
C. IN, ANY, ALL only
D. EXISTS only

Answer: B When the inner query returns exactly one value, the ordinary relational operators can compare it.

Q114 SUBQUERIES

= ANY (subquery) is equivalent to:

A. NOT IN
B. IN
C. = ALL
D. EXISTS

Answer: B = ANY is true when the value equals at least one value in the list, which is exactly what IN means.

Q115 SUBQUERIES

sal > ANY (3000, 2500, 1500, 1200, 1700) is equivalent to:

A. sal > 1500
B. sal > 1200 (greater than the minimum)
C. sal > 3000
D. sal > 1980

Answer: B > ANY is true if sal is greater than at least one value, which only requires it to beat the smallest one.

Q116 SUBQUERIES

weight < ALL (45, 65, 75, 80) is equivalent to:

A. weight < 45 (less than the minimum)
B. weight > 45
C. weight < 80
D. weight < 66.25

Answer: A < ALL must be true for every value, so the value must be smaller than the smallest one.

Q117 SUBQUERIES

<> ALL (subquery) is equivalent to:

A. IN
B. EXISTS
C. = ANY
D. NOT IN

Answer: D <> ALL means 'different from every value in the list', which is NOT IN.

Q118 SUBQUERIES

A correlated subquery is one that:

A. Runs once before the outer query
B. Is written in the FROM clause
C. Always returns more than one row
D. Refers to a column of the outer query and is evaluated for each outer row

Answer: D Example: WHERE sal = (SELECT MAX(sal) FROM emp e2 WHERE e2.deptno = e1.deptno). The inner query depends on the current outer row.

Q119 SUBQUERIES

What happens here?

```
SELECT * FROM emp WHERE sal = (SELECT sal FROM emp WHERE deptno = 10);
```

A. Empty set
B. Returns dept 10 employees
C. Returns the first match
D. ERROR 1242: Subquery returns more than 1 row

Answer: D Dept 10 has three salaries, but = expects a single value. Use IN or = ANY for a multi-row subquery.

Q120 STORED PROGRAMS

Why is DELIMITER $$ used before CREATE PROCEDURE in the mysql client?

A. So that the ; statements inside the body do not end the CREATE PROCEDURE statement early
B. To declare variables
C. To create comments
D. It is required by the server

Answer: A The client normally sends a statement at every ';'. Changing the delimiter lets the whole body reach the server as one statement. DELIMITER is a client command, not SQL.

Q121 STORED PROCEDURES

What is printed?

```
-- CREATE PROCEDURE sqr(IN n INT, OUT r INT)
--   BEGIN SET r = n * n; END
CALL sqr(15, @res);
SELECT @res;
```

A. 225
B. 15
C. 30
D. NULL

Answer: A The OUT parameter r is copied back to the session variable @res when the procedure returns: 15 x 15 = 225.

Q122 STORED PROCEDURES

What is printed?

```
-- CREATE PROCEDURE new_sqr(INOUT n INT)
--   BEGIN SET n = n * n; END
SET @num = 10;
CALL new_sqr(@num);
SELECT @num;
```

A. NULL
B. 1000
C. 100
D. 10

Answer: C INOUT takes the input value (10) and passes the modified value (100) back through the same variable.

Q123 STORED FUNCTIONS

Which statement about stored functions vs stored procedures is TRUE?

A. A procedure must return a value
B. A function must RETURN a single value and can be used in SQL expressions such as SELECT
C. A function can have OUT parameters
D. A procedure can be called inside SELECT

Answer: B Functions return exactly one value (RETURNS type ... RETURN value) and are used in expressions. Procedures are run with CALL and return data through OUT/INOUT parameters or result sets.

Q124 STORED FUNCTIONS

What does SELECT add_num(123.45, NULL); return?

```
CREATE FUNCTION add_num(a DECIMAL(9,2), b DECIMAL(9,2))
RETURNS DECIMAL(9,2) DETERMINISTIC
BEGIN
  RETURN IFNULL(a,0) + IFNULL(b,0);
END
```

A. NULL
B. 0.00
C. 123.45
D. An error

Answer: C IFNULL converts the NULL argument to 0, so the result is 123.45 + 0.

Q125 TRIGGERS

In an AFTER INSERT trigger, which row image is available?

A. NEW only
B. OLD only
C. Neither
D. Both OLD and NEW

Answer: A INSERT creates a row, so there is only NEW. DELETE has only OLD. UPDATE has both.

Q126 LOOPS

What is the difference between WHILE and REPEAT loops in MySQL?

A. No difference
B. REPEAT checks before; WHILE after
C. WHILE cannot be used in procedures
D. WHILE checks the condition before each pass; REPEAT ... UNTIL checks it after each pass, so REPEAT runs at least once

Answer: D REPEAT is an exit-controlled loop like do-while: its body runs at least once and it stops when the UNTIL condition becomes true.

Q127 LOOPS

Which statement exits a labeled LOOP named findmax?

A. BREAK findmax;
B. EXIT findmax;
C. ITERATE findmax;
D. LEAVE findmax;

Answer: D LEAVE label exits the loop (like break). ITERATE label starts the next pass (like continue).

Q128 EXCEPTION HANDLING

What is the difference between an EXIT and a CONTINUE handler?

A. No difference
B. CONTINUE retries the failed statement
C. After the handler runs, EXIT leaves the current BEGIN...END block, while CONTINUE resumes with the statement after the one that failed
D. EXIT ends the session

Answer: C Both run the handler code. Execution then either leaves the enclosing block (EXIT) or carries on with the next statement (CONTINUE).

Q129 EXCEPTION HANDLING

How do you raise a user-defined error such as 'Age should be above 15 yrs.' in a procedure?

A. SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Age should be above 15 yrs.';
B. THROW 45000, 'Age...';
C. RAISE 'Age should be above 15 yrs.';
D. ERROR('Age...');

Answer: A SIGNAL raises an error. SQLSTATE '45000' is the generic state for unhandled user-defined exceptions.

Q130 CURSORS

What is the correct order of steps when using a cursor?

A. OPEN, DECLARE, FETCH, CLOSE

B. DECLARE, FETCH, OPEN, CLOSE

C. DECLARE cursor, DECLARE NOT FOUND handler, OPEN, FETCH in a loop, CLOSE

D. FETCH, OPEN, CLOSE

Answer: C Declare the cursor (after variables), declare the handler that marks the end of data, open it, fetch rows in a loop, then close it.

Q131 CURSORS

Why is DECLARE CONTINUE HANDLER FOR NOT FOUND used with cursors?

A. To avoid declaring variables

B. To open the cursor

C. To ignore duplicate keys

D. To detect that FETCH has no more rows, so the loop can end

Answer: D When FETCH runs past the last row, the NOT FOUND condition (SQLSTATE 02000) is raised. The handler sets a flag that the loop checks before LEAVE.

Q132 TRANSACTIONS

What does ROLLBACK do after START TRANSACTION and a few UPDATEs?

A. Restarts the server

B. Commits them

C. Undoes only the last statement

D. Undoes all DML of the current transaction and ends it

Answer: D ROLLBACK discards every uncommitted change made since START TRANSACTION. COMMIT and ROLLBACK both complete the transaction.

Q133 TRANSACTIONS

What is the default value of @@autocommit in MySQL?

A. 1

B. 0

C. OFF in InnoDB

D. NULL

Answer: A autocommit = 1 means each statement is committed immediately unless you open a transaction with START TRANSACTION (or BEGIN).

Q134 TRANSACTIONS

What does ROLLBACK TO SAVEPOINT s1 do?

A. Rolls back the whole transaction

B. Undoes the changes made after s1; the transaction stays open

C. Deletes s1 and commits

D. Ends the transaction

Answer: B Rolling back to a savepoint is a partial undo. The transaction still needs COMMIT or ROLLBACK to complete.

Q135 DCL

Which command creates a user sunbeam with password 'sunbeam'?

A. CREATE USER sunbeam PASSWORD 'sunbeam';
B. CREATE USER sunbeam IDENTIFIED BY 'sunbeam';
C. ADD USER sunbeam 'sunbeam';
D. GRANT USER sunbeam IDENTIFIED 'sunbeam';

Answer: B CREATE USER 'name'@'host' IDENTIFIED BY 'password'; Without a host, '%' (any host) is used.

Q136 DCL

What does WITH GRANT OPTION add to a GRANT statement?

```
GRANT ALL PRIVILEGES ON ac_classwork_db.* TO mgr WITH GRANT OPTION;
```

A. It makes mgr a root user
B. It removes existing privileges
C. It allows mgr to pass on the privileges it holds to other users
D. It grants privileges on all databases

Answer: C With GRANT OPTION, the grantee can GRANT its own privileges to other users.

Q137 DCL

Which statement removes the UPDATE privilege on emp from dev1?

A. REVOKE UPDATE ON ac_classwork_db.emp TO dev1;
B. REVOKE UPDATE ON ac_classwork_db.emp FROM dev1;
C. DELETE GRANT UPDATE FROM dev1;
D. DROP PRIVILEGE UPDATE dev1;

Answer: B GRANT ... TO user; and REVOKE ... FROM user;

Q138 WINDOW FUNCTIONS

What is the difference between RANK() and DENSE_RANK()?

A. No difference
B. RANK skips rank numbers after ties; DENSE_RANK does not
C. DENSE_RANK skips numbers; RANK does not
D. RANK gives unique numbers even for ties

Answer: B For salaries 5000, 3000, 3000, 2975: RANK gives 1,2,2,4 and DENSE_RANK gives 1,2,2,3. ROW_NUMBER gives 1,2,3,4.

Q139 WINDOW FUNCTIONS

What does PARTITION BY deptno do in SUM(sal) OVER (PARTITION BY deptno)?

A. Groups rows and returns one row per dept
B. Splits the table physically
C. Computes the sum per dept but keeps every row in the output
D. Sorts by deptno

Answer: C Unlike GROUP BY, a window function does not collapse rows. Each employee row shows the total of its own department.

Q140 WINDOW FUNCTIONS

LAG(sal) OVER (ORDER BY sal) returns:

A. The next row's salary
B. The salary rank
C. The lowest salary
D. The previous row's salary (NULL for the first row)

Answer: D LAG looks back and LEAD looks ahead in the ordered window. The default offset is 1 and the default value is NULL.

Q141 CTE

Which is the correct syntax for a Common Table Expression?

A. SELECT * FROM (WITH dept_total) ;
B. WITH dept_total AS (SELECT deptno, SUM(sal) t FROM emp GROUP BY deptno) SELECT AVG(t) FROM dept_total;
C. DECLARE dept_total AS TABLE ...
D. CREATE CTE dept_total AS SELECT ...;

Answer: B WITH name AS (query) defines a temporary named result set used by the statement that follows it.

Q142 NORMALIZATION

A table is in 2NF when it is in 1NF and:

A. No non-key attribute depends on only part of a composite candidate key (no partial dependency)
B. It has no transitive dependency
C. It has no multi-valued dependency
D. Every determinant is a candidate key

Answer: A 2NF removes partial dependencies, which can only occur when the key is composite.

Q143 NORMALIZATION

A table is in 3NF when it is in 2NF and:

A. No non-key attribute depends transitively on the key (non-key depends only on the key)
B. It has a surrogate key
C. It has no NULLs
D. It has only two columns

Answer: A In 3NF, every non-key attribute depends on 'the key, the whole key, and nothing but the key'. A transitive chain such as empno -> deptno -> dname violates it.

Q144 NORMALIZATION

BCNF requires that:

A. All columns are atomic
B. There are no NULL values
C. Every determinant is a candidate key (super key)
D. Every table has a foreign key

Answer: C Boyce-Codd Normal Form is a stricter 3NF: for every non-trivial dependency X -> Y, X must be a super key.

Q145 MONGODB

Which MongoDB query returns employees with sal greater than 2500?

A. db.emp.find({$gt: {sal: 2500}})
B. db.emp.find({sal: {$gt: 2500}})
C. db.emp.find({sal > 2500})
D. db.emp.select({sal: '>2500'})

Answer: B Comparison operators are written as {field: {$op: value}}: $eq, $ne, $gt, $gte, $lt, $lte.

Q146 MONGODB

Which query returns emps sorted by sal in descending order?

A. db.emp.find({sal: -1})
B. db.emp.find().sort({sal: 1})
C. db.emp.find().order({sal: 'desc'})
D. db.emp.find().sort({sal: -1})

Answer: D In sort(), 1 means ascending and -1 means descending.

Q147 MONGODB

Which update operator increments a numeric field by a given amount?

A. $add
B. $inc
C. $push
D. $set

Answer: B {$inc: {age: 1}} adds 1; a negative value subtracts. $set assigns a value and $mul multiplies.

Q148 MONGODB

Which query returns only the documents that have a comm field?

A. db.emp.find({comm: true})
B. db.emp.find({comm: {$exists: 1}})
C. db.emp.find({$has: 'comm'})
D. db.emp.find({comm: {$ne: null}})

Answer: B $exists: true/1 matches documents in which the field is present, whatever its value.

Q149 NOSQL

According to the CAP theorem, a distributed system that experiences a network partition must choose between:

A. Speed and Storage
B. Consistency and Availability
C. Partition and Replication
D. Atomicity and Durability

Answer: B CAP: Consistency, Availability, Partition tolerance. When a partition occurs, a system can guarantee consistency or availability, but not both.

Q150 CODD'S RULES

Which Codd rule says every value is accessible using table name + primary key + column name?

A. Rule 1: Information rule
B. Rule 4: Dynamic online catalog
C. Rule 2: Guaranteed access rule
D. Rule 3: Systematic treatment of NULL

Answer: C Rule 2 guarantees logical access to every atomic value through the table name, the primary key value and the column name.

Difficult 100 questions · Q151–Q250

Q151 NULL

What is the output in MySQL 8?

```
SELECT GREATEST(10, NULL, 30);
```

A. An error
B. 10
C. NULL
D. 30

Answer: C GREATEST() and LEAST() return NULL if any argument is NULL. Wrap arguments in IFNULL/COALESCE to ignore NULLs.

Q152 DATE FUNCTIONS

What does the last comparison return?

```
SELECT NOW() = SYSDATE() FROM (SELECT SLEEP(1)) s;
```

A. 1
B. 0
C. An error
D. NULL

Answer: B NOW() returns the time the statement started and stays constant during it. SYSDATE() returns the actual time at which it is evaluated, which is 1 second later here.

Q153 NULL

How many rows does this return? (KING's mgr is NULL)

```
SELECT ename FROM emp WHERE empno NOT IN (SELECT mgr FROM emp);
```

A. 9
B. 14
C. 8
D. 0

Answer: D The subquery list contains a NULL. x NOT IN (..., NULL) is never TRUE (only FALSE or UNKNOWN), so the result is empty. Add WHERE mgr IS NOT NULL inside the subquery, or use NOT EXISTS.

Q154 NULL

What is the output?

```
SELECT COUNT(*) FROM dept WHERE deptno IN (10, NULL);
SELECT COUNT(*) FROM dept WHERE deptno NOT IN (10, NULL);
```

A. 0 and 0
B. 1 and 3
C. 1 and 4
D. 1 and 0

Answer: D IN (10, NULL) is TRUE for 10. NOT IN (10, NULL) means deptno<>10 AND deptno<>NULL; the second part is UNKNOWN, so no row qualifies.

Q155 DATA TYPES

What does SHOW CREATE TABLE report for col3 after this, and does the INSERT succeed?

```
CREATE TABLE temp(col3 TEXT(4));
INSERT INTO temp VALUES ('abcdefghij');
```

A. text; value truncated to 'abcd'
B. tinytext; INSERT succeeds
C. text(4); INSERT fails with 1406
D. varchar(4); INSERT fails

Answer: B For TEXT(M), MySQL picks the smallest TEXT type that can hold M characters, which is TINYTEXT (up to 255 bytes). The 10-character value fits.

Q156 DDL

Which statement about CREATE TABLE p2 AS SELECT * FROM p; is TRUE?

A. It copies data, PRIMARY KEY and UNIQUE indexes
B. It copies only the structure
C. It fails if p has a primary key
D. It copies data and column definitions (with NOT NULL) but not PRIMARY KEY/UNIQUE/FOREIGN KEY indexes

Answer: D CTAS creates columns from the result set and copies rows. Keys and indexes are not copied. CREATE TABLE p3 LIKE p copies the structure with indexes but no data.

Q157 DDL

What does CREATE TABLE p3 LIKE p; create?

A. An empty table without indexes
B. A view on p
C. An exact copy with data
D. An empty table with the same columns, PRIMARY KEY and indexes

Answer: D LIKE clones the full definition, including indexes, but no rows. Foreign key definitions are not copied.

Q158 REGEXP

How many of ww, wow, woow, wooow, woooow, wooooow, woooooow, wooooooow match '^wo{2,}w$'?

A. 8
B. 7
C. 5
D. 6

Answer: D o{2,} requires at least two o's between the w's. 'ww' (0) and 'wow' (1) fail. woow through wooooooow (2 to 7 o's) give 6 matches.

Q159 REGEXP

Which value matches the pattern 'is$'?

A. isnt
B. Biscuit
C. TOO GOOD
D. this

Answer: D $ anchors to the end of the string. 'this' ends with 'is'. Biscuit contains 'is' in the middle, and isnt starts with it.

Q160 JOINS

emps: Amit(10), Rahul(10), Nilesh(20), Nitin(50), Sarang(50); depts: 10, 20, 30, 40. How many rows does the emulated FULL JOIN return?

```
SELECT e.ename, d.dname FROM emps e LEFT JOIN depts d ON e.deptno = d.deptno
UNION
SELECT e.ename, d.dname FROM emps e RIGHT JOIN depts d ON e.deptno = d.deptno;
```

A. 9
B. 10
C. 5
D. 7

Answer: D The LEFT part gives 5 rows. The RIGHT part adds (NULL, OPS) and (NULL, ACC). The 3 matching rows appear in both halves and UNION removes the duplicates: 5 + 2 = 7.

Q161 JOINS

With the same data, which employees does this return?

```
SELECT e.ename FROM emps e LEFT JOIN depts d ON e.deptno = d.deptno WHERE d.deptno IS NULL;
```

A. Nilesh
B. Nitin, Sarang
C. No rows
D. Amit, Rahul

Answer: B A LEFT JOIN with an IS NULL check on the right table is an anti-join: it finds left rows that have no match. Nitin and Sarang belong to dept 50.

Q162 JOINS

emp_meeting: (100,3),(100,4),(200,1),(200,2),(200,3),(200,4),(200,5),(300,1),(300,2),(300,4). Meeting 200 is in the 'Board Room'. How many rows?

```
SELECT e.ename FROM emps e
 JOIN emp_meeting em ON e.empno = em.empno
 JOIN meeting m ON m.meetno = em.meetno
WHERE m.venue = 'Board Room';
```

A. 10
B. 2
C. 3
D. 5

Answer: D Meeting 200 (Annual meet, Board Room) has 5 attendees: empno 1 to 5.

Q163 SET OPERATORS

emps.deptno = 10,10,20,50,50 and depts.deptno = 10,20,30,40. How many rows does each query return?

```
SELECT deptno FROM emps UNION SELECT deptno FROM depts;
SELECT deptno FROM emps UNION ALL SELECT deptno FROM depts;
```

A. 5 and 5
B. 5 and 9
C. 4 and 9
D. 9 and 9

Answer: B UNION gives the distinct values {10, 20, 50, 30, 40} = 5. UNION ALL keeps all 5 + 4 = 9 rows.

Q164 SET OPERATORS

What happens here?

```
SELECT empno, ename FROM emp
UNION
SELECT deptno FROM dept;
```

A. Returns 18 rows
B. Returns only emp rows
C. ERROR 1222: The used SELECT statements have a different number of columns
D. ename is filled with NULL

Answer: C Every SELECT in a UNION must return the same number of columns, and they should have compatible types. Column names come from the first SELECT.

Q165 INDEXES

A composite index is created on emp(deptno, job). Which WHERE clause can NOT use it efficiently?

A. WHERE deptno = 20 AND job LIKE 'CL%'
B. WHERE deptno = 20
C. WHERE job = 'CLERK'
D. WHERE deptno = 20 AND job = 'CLERK'

Answer: C A B-tree composite index is ordered by its leftmost column first (leftmost-prefix rule). Filtering only on job skips the leading column, so the index cannot be used for a seek.

Q166 SUBQUERIES

What happens when this is run against a copy table emp_copy?

```
DELETE FROM emp_copy
WHERE sal = (SELECT MAX(sal) FROM emp_copy);
```

A. Deletes nothing
B. Deletes all rows
C. ERROR 1093: You can't specify target table 'emp_copy' for update in FROM clause
D. Deletes KING

Answer: C MySQL does not let a DML statement read the table it is modifying in a direct subquery. Wrap it in a derived table: WHERE sal = (SELECT m FROM (SELECT MAX(sal) m FROM emp_copy) x).

Q167 SUBQUERIES

If the subquery returns an empty set, what does the outer query return?

```
SELECT * FROM emp WHERE sal = (SELECT sal FROM emp WHERE ename = 'NOBODY');
```

A. A row of NULLs
B. ERROR 1242
C. Empty set
D. All rows

Answer: C A scalar subquery with no rows evaluates to NULL. sal = NULL is UNKNOWN for every row, so nothing is returned.

Q168 STORED FUNCTIONS

With binary logging enabled, CREATE FUNCTION f1(a INT) RETURNS INT BEGIN RETURN a*2; END fails with ERROR 1418. Which fix is correct?

A. Add DETERMINISTIC (or NO SQL / READS SQL DATA) to the function definition

B. Remove BEGIN...END

C. Use CALL f1()

D. Change RETURNS INT to RETURN INT

Answer: A With binary logging on, a function must declare its nature (DETERMINISTIC, NO SQL or READS SQL DATA), or log_bin_trust_function_creators must be 1.

Q169 TRIGGERS

What happens on CREATE TRIGGER trg AFTER INSERT ON tt FOR EACH ROW SET NEW.x = 5;?

A. ERROR 1362: Updating of NEW row is not allowed in after trigger

B. x is stored as 5

C. The trigger fires twice

D. Works only for the first row

Answer: A In an AFTER trigger the row is already written, so NEW is read-only. Use a BEFORE trigger to change values being inserted.

Q170 TRIGGERS

A BEFORE INSERT trigger on tt does SET NEW.x = NEW.x * 10;. After INSERT INTO tt VALUES (3);, what is stored?

A. 30

B. NULL

C. 3

D. Error: cannot modify NEW

Answer: A A BEFORE trigger may change NEW values before the row is written, so 30 is stored.

Q171 TRIGGERS

A trigger on INSERT refers to OLD.x. What happens?

A. It works only for UPDATE

B. ERROR 1363: There is no OLD row in on INSERT trigger

C. OLD.x is NULL

D. OLD equals NEW

Answer: B INSERT has no previous row image. OLD exists only in UPDATE and DELETE triggers.

Q172 STORED PROGRAMS

Why does this procedure fail to compile?

```
CREATE PROCEDURE d1()
BEGIN
  DECLARE cur CURSOR FOR SELECT x FROM tt;
  DECLARE v INT;
END
```

A. ERROR 1337: variable declaration after cursor or handler declaration

B. Cursors cannot be declared in procedures

C. Missing OPEN

D. v must have a DEFAULT

Answer: A Inside a block, declarations must follow this order: variables and conditions first, then cursors, then handlers.

Q173 CURSORS

What happens when a cursor loop has no NOT FOUND handler and fetches past the last row?

```
LOOP
  FETCH cur INTO v;
END LOOP;
```

A. v becomes NULL forever
B. The loop ends silently
C. ERROR 1329: No data - zero rows fetched, selected, or processed
D. The cursor restarts

Answer: C Without a handler, the NOT FOUND condition becomes an error and the procedure stops.

Q174 EXCEPTION HANDLING

p already contains id 1. What is displayed by CALL h1();?

```
CREATE PROCEDURE h1()
BEGIN
  DECLARE CONTINUE HANDLER FOR 1062 SELECT 'dup' AS m;
  INSERT INTO p VALUES (1, 'z', NULL);
  SELECT 'after' AS m;
END
```

A. Only 'after'
B. ERROR 1062
C. Only 'dup'
D. 'dup' and then 'after'

Answer: D The CONTINUE handler runs ('dup'), then execution continues with the next statement ('after').

Q175 EXCEPTION HANDLING

Same procedure, but with DECLARE EXIT HANDLER FOR 1062. What is displayed?

A. ERROR 1062
B. 'dup' and 'after'
C. Only 'dup'
D. Only 'after'

Answer: C An EXIT handler runs and then leaves the BEGIN...END block, so SELECT 'after' never runs.

Q176 EXCEPTION HANDLING

What is the effect of DECLARE duplicate_entry CONDITION FOR 1062;?

A. It gives a readable name to error code 1062 that can be used in a handler
B. It raises error 1062
C. It creates a table
D. It ignores duplicate entries

Answer: A A named condition is an alias for an error code or SQLSTATE: DECLARE EXIT HANDLER FOR duplicate_entry ...

Q177 EXCEPTION HANDLING

Which handler condition catches every SQL error not caught by a more specific handler?

A. SQLEXCEPTION
B. SQLWARNING
C. 45000
D. NOT FOUND

Answer: A SQLEXCEPTION is shorthand for SQLSTATE values that do not begin with '00', '01' or '02'. NOT FOUND is '02', SQLWARNING is '01'.

Q178 EXCEPTION HANDLING

In a generic SQLEXCEPTION handler, how can you capture the actual error number and message?

A. CATCH(e) e.code
B. SHOW ERRORS INTO v_code;
C. SELECT @@error;
D. GET DIAGNOSTICS CONDITION 1 v_code = MYSQL_ERRNO, v_msg = MESSAGE_TEXT;

Answer: D GET DIAGNOSTICS reads the diagnostics area. The values can then be logged, e.g. INSERT INTO error_log VALUES (v_code, v_msg).

Q179 SELECT

What does this return?

```
SELECT sal FROM emp ORDER BY sal DESC LIMIT 1, 1;
```

A. 3000
B. 5000
C. 2975
D. 2850

Answer: A The descending order is 5000, 3000, 3000, 2975 ... Skipping 1 row returns the second 3000. For the 2nd highest DISTINCT salary use SELECT DISTINCT sal ... LIMIT 1,1, which gives 3000, then 2975 for LIMIT 2,1.

Q180 SELECT

Which query correctly returns the 3rd highest DISTINCT salary (2975)?

A. SELECT MAX(sal) FROM emp LIMIT 3;
B. SELECT DISTINCT sal FROM emp ORDER BY sal DESC LIMIT 2,1;
C. SELECT sal FROM emp ORDER BY sal DESC LIMIT 3;
D. SELECT sal FROM emp ORDER BY sal DESC LIMIT 2,1;

Answer: B Distinct salaries in descending order: 5000, 3000, 2975 ... DISTINCT removes the duplicate 3000, so offset 2 gives 2975. Without DISTINCT, offset 2 returns 3000 again.

Q181 GROUP BY

What is the output?

```
SELECT deptno, COUNT(*) FROM emp GROUP BY deptno HAVING COUNT(*) > 5;
```

A. 10 and 20
B. 20 and 30
C. No rows
D. 30 only

Answer: D Department counts: 10 -> 3, 20 -> 5, 30 -> 6. Only dept 30 has more than 5.

Q182 ROLLUP

How many rows does this query return?

```
SELECT deptno, job, COUNT(*) FROM emp GROUP BY deptno, job WITH ROLLUP;
```

A. 13
B. 9
C. 12
D. 10

Answer: A There are 9 (deptno, job) combinations, 3 dept subtotals and 1 grand total: 9 + 3 + 1 = 13.

Q183 ROLLUP

How many rows does GROUP BY deptno WITH ROLLUP return on emp, and what is in the last row?

```
SELECT deptno, COUNT(*) FROM emp GROUP BY deptno WITH ROLLUP;
```

A. 3 rows; last row 30, 6
B. 4 rows; last row NULL, 14
C. 4 rows; last row 0, 14
D. 3 rows; last row NULL, 14

Answer: B One row per department (3, 5, 6), then the super-aggregate row with deptno NULL and the grand total 14.

Q184 ROLLUP

Which clause keeps ONLY the subtotal and grand-total rows of a deptno, job ROLLUP?

A. HAVING job = 'TOTAL'
B. WHERE GROUPING(deptno) = 1
C. WHERE job IS NULL
D. HAVING GROUPING(job) = 1

Answer: D GROUPING() can only be evaluated after grouping, so it goes in HAVING. GROUPING(job) = 1 marks rows where job was rolled up (subtotals and the grand total).

Q185 AGGREGATE FUNCTIONS

What is the output?

```
SELECT SUM(sal + comm), SUM(sal) + SUM(comm) FROM emp;
```

A. NULL, 31225
B. 7800, 31225
C. 29025, 31225
D. 31225, 31225

Answer: B sal + comm is NULL for the 10 rows without commission, so SUM(sal+comm) only adds the 4 salesmen: (1600+300)+(1250+500)+(1250+1400)+(1500+0) = 7800. The second expression is 29025 + 2200 = 31225.

Q186 AGGREGATE FUNCTIONS

What does this query return?

```
SELECT MAX(AVG(sal)) FROM emp GROUP BY deptno;
```

A. 2916.67
B. 2175
C. ERROR 1111: Invalid use of group function
D. 5000

Answer: C MySQL does not allow nested group functions. Compute the inner aggregate in a derived table or CTE, then apply MAX to it.

Q187 CTE

What is the result?

```
WITH dept_total AS (
  SELECT deptno, SUM(sal) AS t FROM emp GROUP BY deptno
)
SELECT AVG(t) FROM dept_total;
```

A. 29025
B. 8750
C. 2073.21
D. 9675

Answer: D The department totals are 8750, 10875 and 9400. Their average is 29025 / 3 = 9675.

Q188 SUBQUERIES

What happens here?

```
SELECT * FROM (SELECT * FROM emp);
```

A. Infinite loop

B. ERROR 1248: Every derived table must have its own alias

C. Returns all emps

D. Returns empty set

Answer: B A subquery in FROM (derived table) must be given an alias, e.g. (SELECT * FROM emp) AS t.

Q189 SUBQUERIES

How many employees earn the maximum salary of their own department?

```
SELECT ename FROM emp e1
WHERE sal = (SELECT MAX(sal) FROM emp e2 WHERE e2.deptno = e1.deptno);
```

A. 4

B. 5

C. 1

D. 3

Answer: A Dept 10: KING 5000. Dept 20: SCOTT and FORD (both 3000). Dept 30: BLAKE 2850. That is 4 rows.

Q190 SUBQUERIES

How many employees earn more than the average salary of their own department?

```
SELECT ename FROM emp e
WHERE sal > (SELECT AVG(sal) FROM emp WHERE deptno = e.deptno);
```

A. 4

B. 5

C. 6

D. 7

Answer: C Averages: d10 = 2916.67 (KING), d20 = 2175 (JONES, SCOTT, FORD), d30 = 1566.67 (ALLEN, BLAKE). Total 6.

Q191 SUBQUERIES

How many rows does this return?

```
SELECT ename FROM emp
WHERE sal > ALL (SELECT sal FROM emp WHERE job = 'SALESMAN');
```

A. 5

B. 4

C. 10

D. 6

Answer: D > ALL means greater than the highest salesman salary (1600): CLARK, BLAKE, JONES, SCOTT, FORD, KING = 6.

Q192 SUBQUERIES

How many rows does this return?

```
SELECT ename FROM emp
WHERE sal < ANY (SELECT sal FROM emp WHERE deptno = 30);
```

A. 10
B. 9
C. 8
D. 13

Answer: B < ANY means below the maximum of dept 30 (BLAKE 2850). The employees at 2850 or above are BLAKE, JONES, SCOTT, FORD and KING, which leaves 14 - 5 = 9.

Q193 SUBQUERIES

What does this return?

```
SELECT ename FROM emp WHERE sal = ALL (SELECT sal FROM emp WHERE deptno = 10);
```

A. An error
B. KING, CLARK, MILLER
C. Empty set
D. KING

Answer: C = ALL would need sal to equal 5000, 2450 and 1300 at the same time, which is impossible. It only returns rows when the subquery returns one distinct value.

Q194 SUBQUERIES

What does this statement delete?

```
DELETE FROM dept
WHERE deptno NOT IN (SELECT DISTINCT deptno FROM emp);
```

A. Error 1093
B. Dept 40 (OPERATIONS)
C. All depts
D. Nothing, because of NULL

Answer: B emp.deptno contains only 10, 20 and 30 and no NULLs, so NOT IN works and dept 40, which has no employees, is deleted. There is no 1093 error because the subquery reads a different table.

Q195 SUBQUERIES

Which INSERT puts JOHN into the RESEARCH department without hard-coding its number?

A. INSERT INTO emp VALUES (8000, 'JOHN', dept.RESEARCH);
B. INSERT INTO emp(empno, ename, deptno) VALUES (8000, 'JOHN', (SELECT deptno FROM dept WHERE dname = 'RESEARCH'));
C. INSERT INTO emp(empno, ename, deptno) VALUES (8000, 'JOHN', 'RESEARCH');
D. INSERT INTO emp SELECT 'RESEARCH';

Answer: B A scalar subquery can supply a value in the VALUES list, as long as it returns exactly one value.

Q196 JOINS

How many employees fall in salgrade 4 (2001 to 3000)?

```
SELECT e.ename FROM emp e JOIN salgrade s
  ON e.sal BETWEEN s.losal AND s.hisal
WHERE s.grade = 4;
```

A. 5
B. 4
C. 6
D. 3

Answer: A CLARK 2450, BLAKE 2850, JONES 2975, SCOTT 3000 and FORD 3000 = 5.

Q197 JOINS

How many rows do these two self joins return?

```
-- Q1
SELECT e.ename, m.ename FROM emp e JOIN emp m ON e.mgr = m.empno;
-- Q2
SELECT e.ename, m.ename FROM emp e LEFT JOIN emp m ON e.mgr = m.empno;
```

A. 13 and 14
B. 13 and 13
C. 14 and 14
D. 14 and 13

Answer: A KING has no manager (mgr IS NULL), so the inner join drops him (13 rows). The left join keeps him with a NULL manager (14 rows).

Q198 JOINS

Who is SMITH's manager's manager?

```
SELECT m2.ename FROM emp e
 JOIN emp m  ON e.mgr = m.empno
 JOIN emp m2 ON m.mgr = m2.empno
WHERE e.ename = 'SMITH';
```

A. FORD
B. KING
C. JONES
D. BLAKE

Answer: C SMITH -> mgr 7902 (FORD) -> FORD's mgr 7566 (JONES).

Q199 JOINS

emps has 5 rows and meeting has 3 rows. How many rows does SELECT * FROM emps, meeting; return?

A. 8
B. 3
C. 15
D. Error: missing ON

Answer: C Listing tables with commas and no WHERE condition gives a Cartesian product: 5 x 3 = 15.

Q200 WINDOW FUNCTIONS

What are the RANK() and DENSE_RANK() of JONES (2975) when ordered by sal DESC?

```
SELECT ename, RANK() OVER w, DENSE_RANK() OVER w
FROM emp WINDOW w AS (ORDER BY sal DESC);
```

A. 4 and 4
B. 4 and 3
C. 3 and 4
D. 3 and 3

Answer: B KING 5000 is 1; SCOTT and FORD tie at 2. RANK skips 3, so JONES gets 4. DENSE_RANK does not skip, so JONES gets 3.

Q201 WINDOW FUNCTIONS

What is the largest value produced by ROW_NUMBER() OVER (PARTITION BY deptno ORDER BY sal DESC) for dept 20?

A. 14
B. 6
C. 3
D. 5

Answer: D ROW_NUMBER restarts at 1 in each partition. Dept 20 has 5 employees, so the numbers go from 1 to 5.

Q202 WINDOW FUNCTIONS

What running total is shown on WARD's row (sal 1250)?

```
SELECT ename, sal, SUM(sal) OVER (ORDER BY sal) FROM emp;
```

A. 2850
B. 5350
C. 4100
D. 3100

Answer: B With ORDER BY and no frame, the default frame is RANGE ... CURRENT ROW, which includes peer rows with the same sal. WARD and MARTIN (both 1250) therefore both show 800+950+1100+1250+1250 = 5350.

Q203 WINDOW FUNCTIONS

For dept 10 ordered by sal DESC, what does LAST_VALUE(ename) show on CLARK's row?

```
SELECT ename,
  FIRST_VALUE(ename) OVER (PARTITION BY deptno ORDER BY sal DESC) AS fv,
  LAST_VALUE(ename)  OVER (PARTITION BY deptno ORDER BY sal DESC) AS lv
FROM emp WHERE deptno = 10;
```

A. CLARK
B. NULL
C. KING
D. MILLER

Answer: A The default frame ends at the current row, so LAST_VALUE returns the current row's name (CLARK). Use ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING to get MILLER.

Q204 WINDOW FUNCTIONS

What do LAG(sal) and LEAD(sal, 1, 0) show on SMITH's row (the lowest salary) when ordered by sal?

```
SELECT ename, LAG(sal) OVER (ORDER BY sal), LEAD(sal,1,0) OVER (ORDER BY sal) FROM emp;
```

A. 0 and 950
B. 800 and 950
C. NULL and 950
D. NULL and 0

Answer: C SMITH is the first row, so there is no previous row and LAG returns NULL (no default given). LEAD returns the next salary, 950 (JAMES).

Q205 WINDOW FUNCTIONS

What value appears on every row?

```
SELECT ename, sal, SUM(sal) OVER () AS total FROM emp;
```

A. 29025
B. Error: OVER needs PARTITION BY
C. A running total
D. The employee's salary

Answer: A An empty OVER() makes the whole result one window, so each row shows the company total of 29025.

Q206 WINDOW FUNCTIONS

Where can a window function such as RANK() OVER (...) NOT be used directly?

A. In the WHERE clause
B. In the SELECT list
C. Inside a CTE's SELECT list
D. In ORDER BY

Answer: A Window functions are evaluated after WHERE, GROUP BY and HAVING. To filter on them (e.g. rank = 1), compute them in a derived table or CTE and filter outside.

Q207 WINDOW FUNCTIONS

Which query lists the top earner of each department using a window function?

A. SELECT * FROM (SELECT e.*, RANK() OVER (PARTITION BY deptno ORDER BY sal DESC) r FROM emp e) t WHERE r = 1;
B. SELECT deptno, RANK(sal) FROM emp GROUP BY deptno;
C. SELECT * FROM emp WHERE RANK() OVER (PARTITION BY deptno) = 1;
D. SELECT * FROM emp QUALIFY rank = 1;

Answer: A Rank inside each partition in a derived table, then filter r = 1 outside. RANK keeps ties, so SCOTT and FORD both appear for dept 20.

Q208 RECURSIVE CTE

Which query prints 1 to 5?

A. WITH RECURSIVE n AS (SELECT x+1 FROM n WHERE x < 5) SELECT * FROM n;
B. WITH RECURSIVE n AS (SELECT 1 UNION SELECT 5) SELECT * FROM n;
C. WITH n AS (SELECT 1 x UNION ALL SELECT x+1 FROM n) SELECT * FROM n;
D. WITH RECURSIVE n AS (SELECT 1 x UNION ALL SELECT x+1 FROM n WHERE x < 5) SELECT * FROM n;

Answer: D A recursive CTE needs the RECURSIVE keyword, an anchor member (SELECT 1), a recursive member that refers to the CTE, and a stop condition (WHERE x < 5).

Q209 RECURSIVE CTE

What happens if the recursive member has no terminating condition?

A. It returns the anchor row only

B. It runs forever and crashes the server

C. It stops after cte_max_recursion_depth (1000 by default) with an error

D. It returns NULL

Answer: C MySQL limits recursion with cte_max_recursion_depth (default 1000) and raises an error once that depth is passed.

Q210 RECURSIVE CTE

Employees were hired only in 1980, 1981, 1982 and 1983. How many years from 1975 to 1985 have no hires?

```
WITH RECURSIVE y AS (SELECT 1975 yr UNION ALL SELECT yr+1 FROM y WHERE yr < 1985)
SELECT yr FROM y WHERE yr NOT IN (SELECT YEAR(hire) FROM emp);
```

A. 8

B. 11

C. 6

D. 7

Answer: D There are 11 years from 1975 to 1985. Removing 1980 to 1983 leaves 1975-1979 and 1984-1985 = 7.

Q211 RECURSIVE CTE

In the employee-hierarchy CTE (KING at level 1), what level is SMITH?

```
WITH RECURSIVE h AS (
  SELECT empno, ename, 1 AS lvl FROM emp WHERE mgr IS NULL
  UNION ALL
  SELECT e.empno, e.ename, h.lvl + 1 FROM emp e JOIN h ON e.mgr = h.empno
) SELECT * FROM h;
```

A. 4

B. 5

C. 3

D. 2

Answer: A KING (1) -> JONES (2) -> FORD (3) -> SMITH (4).

Q212 RECURSIVE CTE

In the same hierarchy, which employees are at level 2?

A. JONES, BLAKE, CLARK

B. KING

C. SCOTT, FORD

D. ALLEN, WARD

Answer: A Level 2 is everyone who reports directly to KING (7839): JONES, BLAKE and CLARK.

Q213 TRANSACTIONS

What is in ck after this sequence (ck was empty)?

```
START TRANSACTION;
INSERT INTO ck VALUES (20);
CREATE TABLE junk(x INT);
ROLLBACK;
SELECT COUNT(*) FROM ck;
```

A. Depends on autocommit
B. 0
C. Error
D. 1

Answer: D A DDL statement causes an implicit COMMIT, which ends the transaction and saves the pending INSERT. The later ROLLBACK has nothing to undo.

Q214 TRANSACTIONS

What does ck contain after this (it held 20 before)?

```
START TRANSACTION;
INSERT INTO ck VALUES (30);
SAVEPOINT s1;
INSERT INTO ck VALUES (40);
ROLLBACK TO s1;
COMMIT;
```

A. 20, 30, 40
B. 20, 40
C. 20, 30
D. 20

Answer: C ROLLBACK TO s1 undoes only the work done after the savepoint (40). The transaction is still open, and COMMIT saves 30.

Q215 TRANSACTIONS

ck has 2 rows. What does COUNT(*) return at the end?

```
START TRANSACTION;
TRUNCATE ck;
ROLLBACK;
SELECT COUNT(*) FROM ck;
```

A. 2
B. 1
C. Error
D. 0

Answer: D TRUNCATE is DDL. It commits implicitly and cannot be rolled back, so the table stays empty.

Q216 TRANSACTIONS

With autocommit = 1 and NO START TRANSACTION, you run UPDATE accounts SET balance = 0; then ROLLBACK;. What happens?

A. The update stays, because it was already auto-committed
B. ROLLBACK raises an error
C. The update is undone
D. Only half the rows are restored

Answer: A Each statement outside an explicit transaction is committed as soon as it finishes. ROLLBACK can only undo DML run inside a transaction.

Q217 TRANSACTIONS

After SET autocommit = 0; INSERT ...; the user exits the session without COMMIT. What happens to the insert?

A. It is rolled back
B. It stays pending for the next login
C. It is committed automatically on exit
D. The table is locked forever

Answer: A With autocommit off, a transaction is always open. Ending the session without COMMIT rolls back the uncommitted work.

Q218 TRANSACTIONS

What is the default transaction isolation level of InnoDB in MySQL 8?

A. SERIALIZABLE
B. READ UNCOMMITTED
C. REPEATABLE READ
D. READ COMMITTED

Answer: C @@transaction_isolation = 'REPEATABLE-READ'. A transaction sees a consistent snapshot, so repeated reads return the same data.

Q219 TRANSACTIONS

Reading another transaction's uncommitted changes is called a:

A. Dirty read
B. Phantom read
C. Non-repeatable read
D. Lost update

Answer: A A dirty read sees uncommitted data that may later be rolled back. Only READ UNCOMMITTED allows it.

Q220 TRANSACTIONS

Which ACID property guarantees that committed data survives a crash or power failure?

A. Isolation
B. Durability
C. Consistency
D. Atomicity

Answer: B Durability: once COMMIT returns, the change is permanent. InnoDB ensures this with its redo log.

Q221 LOCKING

What does SELECT * FROM accounts WHERE id = 1 FOR UPDATE; do inside a transaction?

A. Updates the row
B. Locks the whole database
C. Places an exclusive row lock until COMMIT/ROLLBACK (pessimistic locking)
D. Nothing, because FOR UPDATE is ignored

Answer: C FOR UPDATE locks the selected rows so other transactions cannot modify (or lock) them until this one ends. This is pessimistic locking.

Q222 LOCKING

Optimistic locking is usually implemented by:

A. LOCK TABLES ... WRITE

B. Keeping a version/timestamp column and updating only if it has not changed

C. SELECT ... FOR UPDATE

D. Setting autocommit = 0

Answer: B Optimistic locking takes no lock while reading. The UPDATE ... WHERE id = ? AND version = ? fails if another transaction changed the row first.

Q223 LOCKING

Session A runs LOCK TABLES emp READ;. What can session B do on emp?

A. Read and write

B. Read but not write (writes wait until UNLOCK TABLES)

C. Only DDL

D. Neither read nor write

Answer: B A READ lock is shared: everyone can read, nobody can write. A WRITE lock is exclusive to the session that holds it.

Q224 DCL

dev1 was granted SELECT, INSERT, UPDATE on emp, then REVOKE UPDATE ON emp FROM dev1; was run. What can dev1 do on emp now?

A. SELECT, INSERT and UPDATE

B. SELECT and INSERT

C. Nothing

D. SELECT only

Answer: B REVOKE removes only the privilege named. SELECT and INSERT remain.

Q225 DCL

An intern has SELECT on view v1_emp but no privilege on the base table emp. What happens with SELECT * FROM v1_emp;?

A. Access denied

B. Only column names are shown

C. It works, because the view (SQL SECURITY DEFINER by default) runs with its creator's privileges

D. The view is dropped

Answer: C Granting on a view while withholding the base table is a common way to restrict users to certain rows or columns.

Q226 NORMALIZATION

Table: Result(student_id, course_id, student_name, grade) with key (student_id, course_id). Which normal form does it violate?

A. BCNF only

B. 2NF, because student_name depends only on student_id (partial dependency)

C. 1NF

D. 3NF only

Answer: B student_name depends on part of the composite key. Move it to Student(student_id, student_name).

Q227 NORMALIZATION

Table: Emp(empno, ename, deptno, dname) with key empno. Which dependency violates 3NF?

A. ename -> empno
B. empno -> ename
C. deptno -> dname (transitive: empno -> deptno -> dname)
D. empno -> deptno

Answer: C dname depends on a non-key attribute (deptno), not directly on the key. Split it into Dept(deptno, dname).

Q228 NORMALIZATION

R(A, B, C) has FDs AB -> C and C -> B. The candidate keys are AB and AC. In which normal form is R?

A. 3NF but not BCNF
B. 1NF only
C. Not even 2NF
D. BCNF

Answer: A C -> B: C is not a super key, so BCNF fails. B is a prime attribute (part of key AB), so 3NF is still satisfied.

Q229 NORMALIZATION

R(A, B, C, D) has FDs A -> B, B -> C, C -> D. What is the candidate key and the closure A+?

A. Key D; A+ = A
B. Key A; A+ = ABCD
C. Key AB; A+ = AB
D. Key ABCD; A+ = ABC

Answer: B A -> B -> C -> D, so A determines every attribute (A+ = ABCD) and A is the candidate key. R is in 2NF but not 3NF because of the transitive chain.

Q230 NORMALIZATION

A customer's address is stored in 50 order rows. Changing it in only some rows leaves inconsistent data. This is a(n):

A. Referential anomaly
B. Insertion anomaly
C. Deletion anomaly
D. Update anomaly

Answer: D Redundant copies must all be updated together. Normalization stores the address once in a Customer table.

Q231 CODD'S RULES

Codd's Rule 4 (dynamic online catalog) requires that:

A. Views are updatable
B. NULLs are supported
C. Data is encrypted
D. Metadata (table and column definitions) is stored in tables and can be queried with the same language, e.g. information_schema

Answer: D In MySQL, information_schema exposes the catalog as ordinary tables that SQL can query.

Q232 CODD'S RULES

How many rules did E. F. Codd define, and what is the first one called?

A. 10 rules; Access rule
B. 13 rules (0 to 12); Rule 0: Foundation rule
C. 12 rules; Foundation rule
D. 12 rules; Information rule

Answer: B Rule 0 (Foundation) says that a relational DBMS must manage databases entirely through its relational capabilities. Rules 1 to 12 elaborate on it.

Q233 CODD'S RULES

Codd's Rule 3 says NULL must be:

A. Treated as 0 for numbers
B. Avoided in all tables
C. A systematic marker for missing/inapplicable data, distinct from 0 and ''
D. Treated as an empty string for text

Answer: C NULL is not 0 and not ''. It must be handled the same way for every data type.

Q234 MONGODB

What happens with this query?

```
db.emp.find({}, {_id: 1, job: 0, sal: 1, comm: 0})
```

A. Shows all fields except job and comm
B. Shows _id and sal
C. Error: Cannot do inclusion on field sal in exclusion projection
D. Shows only _id

Answer: C A projection must be all inclusion (1) or all exclusion (0). The only exception is _id, which may be excluded in an inclusion projection.

Q235 MONGODB

Which employee does this return?

```
db.emp.find().sort({sal: 1}).skip(2).limit(1)
```

A. ADAMS
B. SMITH
C. JAMES
D. WARD

Answer: A Ascending salaries: SMITH 800, JAMES 950, ADAMS 1100 ... skip(2) moves past two documents and limit(1) returns ADAMS (the 3rd lowest). It is the same as LIMIT 2,1 in SQL.

Q236 MONGODB

How many documents does this count? (ALLEN, WARD, MARTIN and TURNER were inserted with a comm field; TURNER's comm is 0)

```
db.emp.find({comm: {$exists: 1}}).count()
```

A. 10
B. 14
C. 4
D. 3

Answer: C $exists checks whether the field is present, not its value. TURNER's comm: 0 counts, so the result is 4.

Q237 MONGODB

What does this count?

```
db.emp.find({sal: {$gt: 2500}}).count()
```

A. 3
B. 4
C. 5
D. 6

Answer: C JONES 2975, BLAKE 2850, SCOTT 3000, KING 5000 and FORD 3000 = 5.

Q238 MONGODB

Which names match this query?

```
db.emp.find({ename: /^[A-Z]{4}$/i})
```

A. No names, because of the i flag
B. Names of exactly 4 letters: WARD, KING, FORD
C. Names containing 4 letters anywhere
D. Names starting with 4 capital letters

Answer: B ^ and $ anchor the whole value, so exactly 4 letters are required. The i flag makes the match case-insensitive.

Q239 MONGODB

What does this do?

```
db.emp.updateMany({job: 'CLERK'}, {$mul: {sal: 1.2}})
```

A. Updates only the first clerk
B. Adds 1.2 to sal
C. Sets every clerk's sal to 1.2
D. Increases every clerk's sal by 20%

Answer: D $mul multiplies the field value. updateMany applies the change to every matching document; updateOne would change only the first.

Q240 MONGODB

What does {upsert: true} do here when no document has name 'Ram'?

```
db.person.updateOne({name: 'Ram'}, {$set: {age: 25}}, {upsert: true})
```

A. Does nothing
B. Raises an error
C. Updates every document
D. Inserts a new document {name: 'Ram', age: 25}

Answer: D Upsert = update if a match exists, otherwise insert. The equality fields of the criteria are included in the new document.

Q241 NOSQL

BASE, the consistency model often followed by NoSQL systems, stands for:

A. Balanced, Available, Scalable, Efficient
B. Basically Available, Soft state, Eventually consistent
C. Binary Access Storage Engine
D. Basic Atomic Serial Execution

Answer: B BASE is the counterpart of ACID: it prefers availability and reaches consistency over time rather than immediately.

Q242 NOSQL

Which database-to-NoSQL-type pairing is INCORRECT?

A. Neo4j: Document store
B. Redis: Key-value store
C. Cassandra: Column-family store
D. MongoDB: Document store

Answer: A Neo4j is a graph database (nodes and relationships). The other three pairings are correct.

Q243 STORED PROCEDURES

What does CALL sum_even(1, 10); display?

```
CREATE PROCEDURE sum_even(v_low INT, v_high INT)
BEGIN
  DECLARE v_i INT DEFAULT v_low;
  DECLARE v_sum INT DEFAULT 0;
  WHILE v_i <= v_high DO
    IF v_i % 2 = 0 THEN SET v_sum = v_sum + v_i; END IF;
    SET v_i = v_i + 1;
  END WHILE;
  SELECT v_sum AS result;
END
```

A. 55
B. 20
C. 30
D. 25

Answer: C The even numbers from 1 to 10 are 2, 4, 6, 8 and 10, which add up to 30.

Q244 STORED PROCEDURES

How many rows does CALL num_table(5); insert into result?

```
CREATE PROCEDURE num_table(v_num INT)
BEGIN
  DECLARE v_i INT DEFAULT 1;
  REPEAT
    INSERT INTO result VALUES (v_num, CONCAT(v_num,' * ',v_i,' = ',v_num*v_i));
    SET v_i = v_i + 1;
  UNTIL v_i > 10
  END REPEAT;
END
```

A. 11
B. 1
C. 9
D. 10

Answer: D The body runs for v_i = 1 to 10. After the 10th insert v_i becomes 11, the UNTIL condition is true and the loop stops.

Q245 CURSORS

t1 holds A, B, C, D and t2 holds B, Z, A, C (in that fetch order). The procedure compares the rows pair by pair and stores the larger one. Which values are stored?

```
FETCH v_cur1 INTO v1;
IF v_err = 1 THEN LEAVE findmax; END IF;
FETCH v_cur2 INTO v2;
IF v1 > v2 THEN INSERT ... v1 ELSE INSERT ... v2 END IF;
```

A. B, Z, A, C
B. B, Z, C, D
C. Z, Z, Z, Z
D. A, B, C, D

Answer: B Pairs: (A,B) -> B, (B,Z) -> Z, (C,A) -> C, (D,C) -> D. The fifth fetch on t1 raises NOT FOUND, the flag is set and the loop is left.

Q246 TRIGGERS

accounts(id 1) has balance 10000. The AFTER INSERT trigger on transactions subtracts the amount for 'WITHDRAW' and adds it otherwise. What is the balance after these two inserts?

```
INSERT INTO transactions VALUES (1, 'WITHDRAW', NOW(), 2000);
INSERT INTO transactions VALUES (1, 'DEPOSIT',  NOW(), 5000);
```

A. 17000
B. 13000
C. 10000
D. 8000

Answer: B 10000 - 2000 = 8000, then 8000 + 5000 = 13000. The trigger runs once for each inserted row.

Q247 TRIGGERS

Which statement is NOT allowed inside a MySQL trigger body?

A. SET NEW.col = value (in a BEFORE trigger)
B. IF ... THEN ... END IF
C. COMMIT or START TRANSACTION
D. UPDATE on another table

Answer: C A trigger runs as part of the statement that fired it, so it cannot begin or end transactions (explicit or implicit commits are not allowed).

Q248 TRIGGERS

Can a table in MySQL 8 have two BEFORE INSERT triggers?

A. Yes; FOLLOWS/PRECEDES control their order
B. Only on MyISAM tables
C. No, only one trigger per table
D. Only if one is AFTER

Answer: A Since MySQL 5.7.2, several triggers with the same timing and event are allowed. By default they run in creation order; FOLLOWS or PRECEDES changes the order.

Q249 STRING FUNCTIONS

What is the output?

```
SELECT LPAD('abc', 2, '*'), CONCAT('a', NULL), CONCAT_WS('-', 'a', NULL, 'b');
```

A. ab, NULL, a-b
B. *abc, a, a--b
C. abc, NULL, a-NULL-b
D. **, a, a-b

Answer: A LPAD truncates when the target length is shorter than the string ('ab'). CONCAT returns NULL if any argument is NULL. CONCAT_WS skips NULL arguments.

Q250 NUMERIC FUNCTIONS

What is the output?

```
SELECT 10/0, 5 DIV 2, MOD(-7, 3), ROUND(-2.5);
```

A. Infinity, 2, -1, -3
B. Error, 2.5, 2, -2
C. NULL, 2, 2, -2
D. NULL, 2, -1, -3

Answer: D Division by zero gives NULL (with a warning). DIV is integer division. MOD takes the sign of the dividend, so -7 mod 3 = -1. ROUND on an exact value rounds half away from zero, so -2.5 becomes -3.

Answer key

Easy

1 C	2 C	3 D	4 A	5 D	6 B	7 B	8 C	9 D	10 B
11 D	12 C	13 A	14 D	15 D	16 B	17 B	18 B	19 C	20 A
21 A	22 C	23 C	24 B	25 B	26 A	27 A	28 C	29 D	30 A
31 B	32 B	33 C	34 D	35 D	36 A	37 A	38 A	39 A	40 B
41 D	42 A	43 D	44 B	45 B	46 C	47 D	48 C	49 C	50 A

Moderate

51 C	52 A	53 A	54 D	55 D	56 B	57 B	58 B	59 C	60 C
61 B	62 A	63 C	64 C	65 A	66 C	67 C	68 D	69 C	70 B
71 D	72 A	73 B	74 A	75 D	76 D	77 A	78 A	79 A	80 D
81 B	82 C	83 D	84 C	85 A	86 A	87 C	88 C	89 A	90 D
91 B	92 D	93 A	94 A	95 D	96 C	97 B	98 C	99 C	100 C
101 D	102 A	103 C	104 B	105 B	106 D	107 A	108 A	109 D	110 B
111 D	112 D	113 B	114 B	115 B	116 A	117 D	118 D	119 D	120 A
121 A	122 C	123 B	124 C	125 A	126 D	127 D	128 C	129 A	130 C
131 D	132 D	133 A	134 B	135 B	136 C	137 B	138 B	139 C	140 D
141 B	142 A	143 A	144 C	145 B	146 D	147 B	148 B	149 B	150 C

Difficult

151 C	152 B	153 D	154 D	155 B	156 D	157 D	158 D	159 D	160 D
161 B	162 D	163 B	164 C	165 C	166 C	167 C	168 A	169 A	170 A
171 B	172 A	173 C	174 D	175 C	176 A	177 A	178 D	179 A	180 B
181 D	182 A	183 B	184 D	185 B	186 C	187 D	188 B	189 A	190 C
191 D	192 B	193 C	194 B	195 B	196 A	197 A	198 C	199 C	200 B
201 D	202 B	203 A	204 C	205 A	206 A	207 A	208 D	209 C	210 D
211 A	212 A	213 D	214 C	215 D	216 A	217 A	218 C	219 A	220 B
221 C	222 B	223 B	224 B	225 C	226 B	227 C	228 A	229 B	230 D
231 D	232 B	233 C	234 C	235 A	236 C	237 C	238 B	239 D	240 D
241 B	242 A	243 C	244 D	245 B	246 B	247 C	248 A	249 A	250 D